ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

**Mestrado em Engenharia Informática – Cibersegurança e Administração de Sistemas.**

# ASSIGNMENT 1

APRAU

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# 1 Abstract

This study uses exploratory data analysis and supervised machine learning techniques with a cardiometabolic data set for **2,000 patients** supplied to **Group 26**. These comprise the prediction of continuous target variable (target_regression) and classification into one of the three diagnostic classes (target_class). We compare simple and multiple linear **regression**, class-balanced **Logistic Regression** and **LDA** with equal class priors, five different resampling strategies (holdout, 5-fold, and 10-fold cross-validation, leave-one-out and bootstrap) as well as L1/L2-regularized logistic models for feature selection. All the model selections are done using cross-validation of the training subset, while the final assessment involves a stratified 20% holdout.

The regression problem shows virtually no linear predictive power. The best performing model, which relies on **BMI** and **HbA1c**, has a holdout **$R^2$ of 0.0037** and **RMSE of 38.87**. The classification problem is extremely imbalanced (64.35%, 32.20% and 3.45%). Large differences in within-class variance were also observed during the exploratory data analysis, especially for **HT_HY**, calling into question the assumption of shared covariances that is used in LDA. The uniform prior **LDA** had the highest value of macro-F1 during 5-fold training (0.5997 compared to Logistic Regression 0.5768) and hold-out macro-F1 equal to 0.5240, however, both classifiers performed equally well during full LOOCV, 10-fold CV and bootstrap. The best performance of L2 regularized model in terms of macro-F1 score during training was observed for the whole model with C estimated via cross-validation (0.5848).

**Keywords:** exploratory data analysis; linear regression; logistic regression; linear discriminant analysis; class imbalance; bootstrap; regularization; feature selection.

# 2 Introduction and Research Objectives

In this assignment, APRAU, we need to analyze the patient-level clinical and lifestyle variables with the help of several machine learning algorithms. There is a dataset consisting of demographic factors, anthropometric indices, laboratory tests, and behavioral variables along with two targets. In this study, we will be comparing the machine learning methods through an open and replicable experiment and will assess their generalization abilities [1].

There are four questions that we need to answer:

1. What are the features and relations of the dataset of Group 26?
2. Can one predictor or a combination of several predictors predict target_regression precisely?
3. How does Logistic Regression perform in comparison to LDA in an imbalanced three-class problem, how the resampling impacts the stability of results?
4. Can sparse regularizer decrease the complexity of the model and improve classification?

Class labels (DI, DI_HY, and HT_HY) and the units of measure of target_regression were reproduced as they were stated in the initial dataset; no conclusions about clinical meanings and measurements were made on our part.

# 3 Dataset and Exploratory Data Analysis

## 3.1 Approach and data integrity

The initial exploration was carried out by investigating the dimensions of the dataset, the types of data, missing values, duplicated records, univariate distributions, and the proportion of classes for the diagnostic class. Descriptive statistics of mean, standard deviation, and quantiles were computed for numerical attributes, whereas descriptive statistics of counts were computed for categorical attributes. Linear relationships of the regression target were investigated through pearson correlations, while exploratory chi-square tests were carried out for classification patterns through class-conditioned boxplots.

***Table 1 - Approach and Data Integrity***

| Data Property | Observed Result |
|---|---|
| **Records / columns** | 2,000 / 35 |
| **Original predictors / targets** | 33 / 2 |
| **Column types** | 22 numeric (21 float, 1 integer); 13 text |
| **Missing values / exact duplicates** | 0 / 0 |
| **Regression target** | Mean 138.192; SD 42.366; range 70–300 |

## 3.2 Diagnostic class imbalance

***Table 2 - Diagnostic class imbalance***

| Class | Patients | Share |
|---|---|---|
| **DI_HY** | 1,287 | 64.35% |
| **DI** | 644 | 32.20% |
| **HT_HY** | 69 | 3.45% |

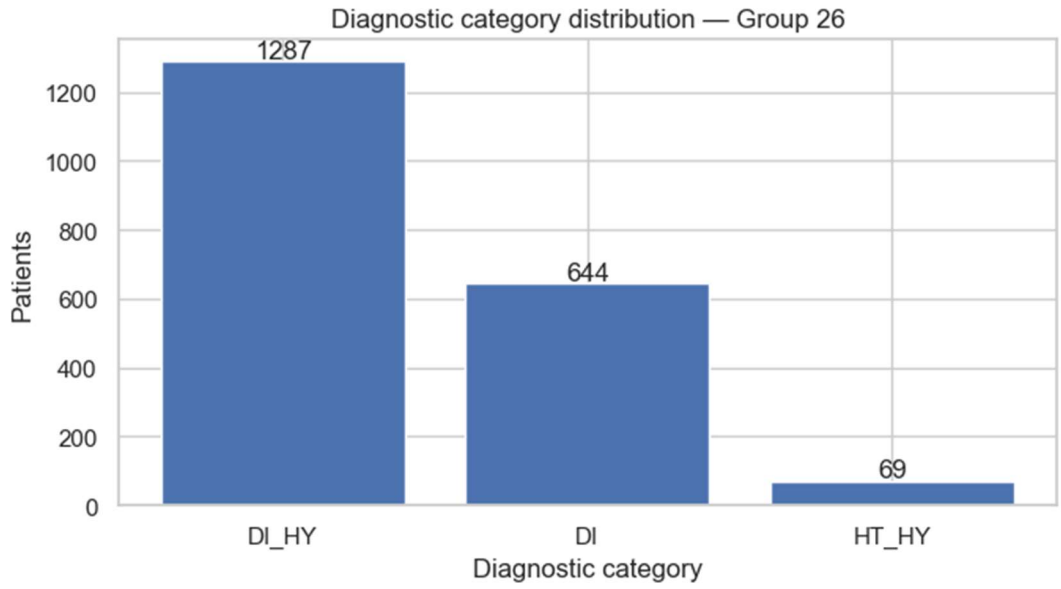



***Figure 1 - Diagnostic target frequencies in Group 26 (notebook cell 5).***

Diagnostic classes are heavily unbalanced. A model that would always predict DI_HY class obtains 64.35% accuracy on the whole dataset without finding any instances of DI and HT_HY. For these reasons, we apply macro-F1, balanced accuracy, and recall for each class as the key metrics. In addition, HT_HY has a small number of samples, making hold-out estimation error-sensitive.

## 3.3 Numerical distributions and associations

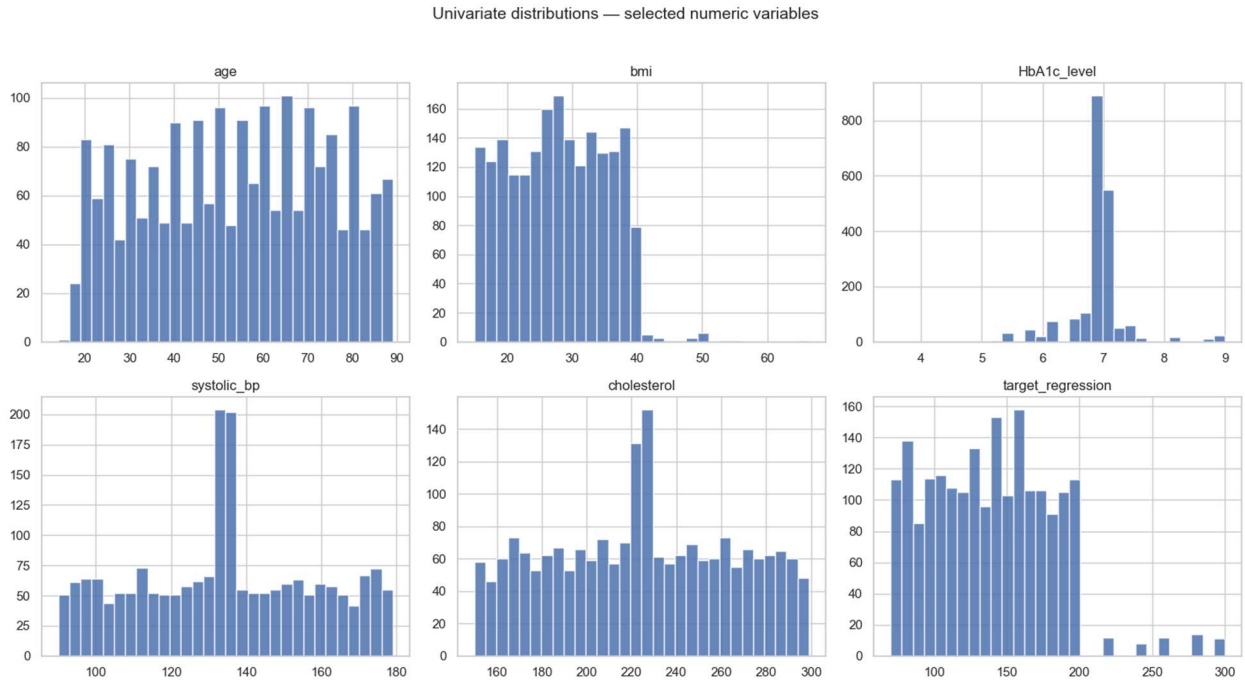



***Figure 2 - Selected numeric distributions, including the continuous response (notebook cell 7).***

Mean age is 53.681 years (sd 20.139) and mean BMI is 27.891 (sd 7.429). Target_regression is characterized by a mean of 138.192 and sd of 42.366. The Pearson correlation of BMI with the target variable is highest (r = +0.0694); HbA1c comes in second place (r = +0.0620) and triglycerides (r = +0.0479) third. All the above correlations are weak linear relations, which means that even if the regression model fits the data well, it still may not explain much. Correlation does not imply causality.

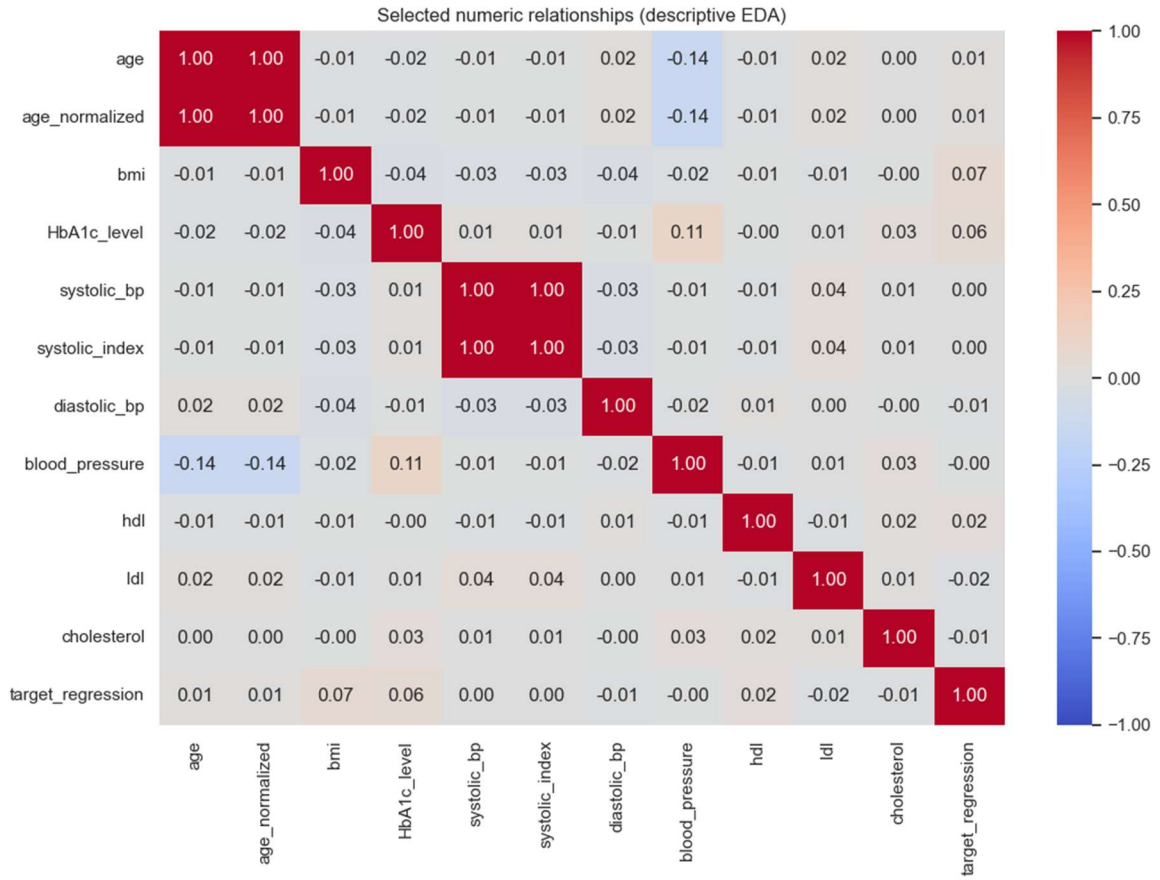



***Figure 3 - Correlations between selected numeric measurements and the regression target (notebook cell 8).***

Chi-square associations of diagnosis revealed the following significant factors: **low_hdl_cholesterol** (p-value < 0.00001 from the rounded notebook table), **smoking** (p = 0.00019), **sugar_consumption** (p = 0.00484) and **education_level** (p = 0.01033). The mentioned p-values are unadjusted and are only indicative in terms of multiple comparisons, without implying any kind of advantage or relationship. In regard to mean HbA1c levels across the classes: 6.91 in DI, 6.95 in DI_HY, and 5.52 in HT_HY, the last one being less common.

## 3.4 Categorical predictors and target_regression

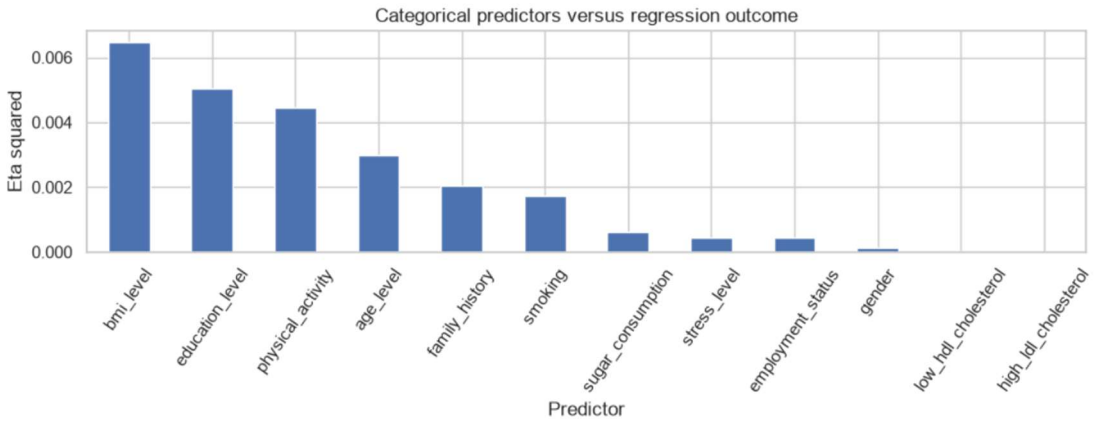



***Figure 4 - Categorical predictors versus the continuous regression target (notebook cell 10)***

***Table 3 - Categorical predictors versus the continuous regression target results***

| Categorical predictor | Eta-squared ($\eta^2$) |
|---|---|
| bmi_level | 0.00648 |
| education_level | 0.00505 |
| physical_activity | 0.00444 |
| age_level | 0.00297 |
| family_history | 0.00205 |
| smoking | 0.00172 |

The relationship between categorical independent variables and the dependent variable was measured using eta squared ($\eta^2$), which is defined as the proportion of variance in the dependent variable that can be attributed to differences in the means of the categories. All values calculated were less than 0.01. The strongest relationship was observed for bmi_level ($\eta^2 = 0.00648$), followed by education_level (0.00505) and physical_activity (0.00444).

## 3.5 Within-class variability and possible data-quality artefacts

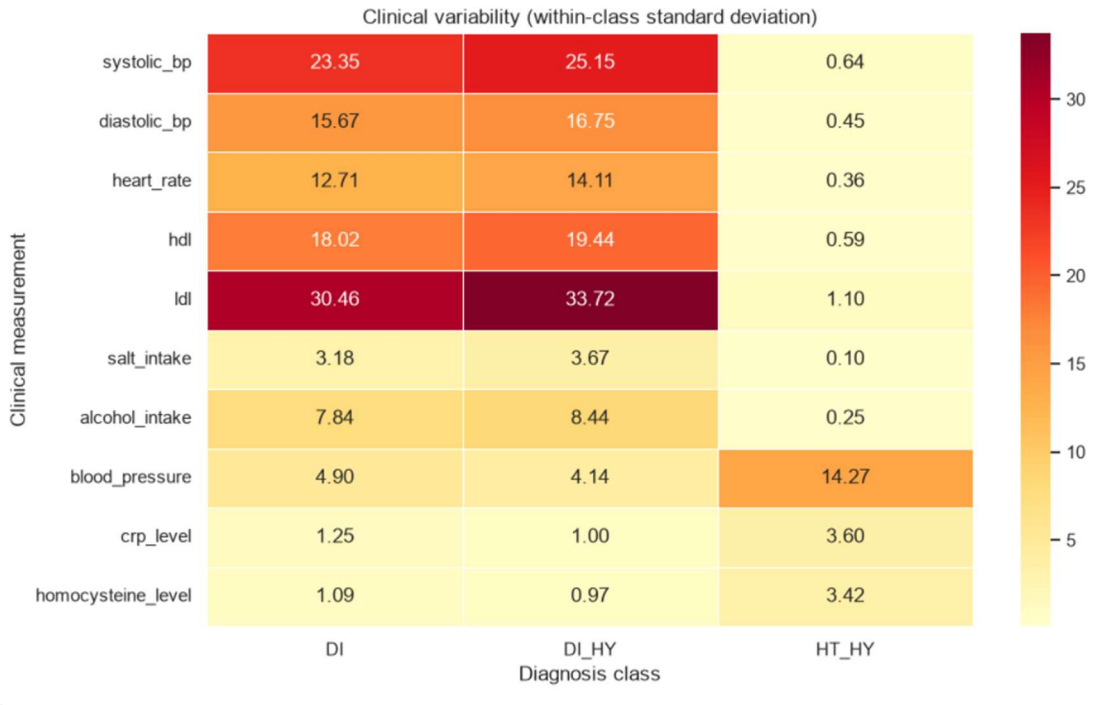



***Figure 5 - Unusual within-class variability and possible data-source artefacts (notebook cell 13)***

***Table 4 - Unusual within-class variability and possible data-source artefacts results***

| Clinical feature | SD DI | SD DI_HY | SD HT_HY |
|---|---|---|---|
| **systolic_bp** | 23.35 | 25.15 | 0.64 |
| **diastolic_bp** | 15.67 | 16.75 | 0.45 |
| **heart_rate** | 12.71 | 14.11 | 0.36 |
| **hdl** | 18.02 | 19.44 | 0.59 |
| **ldl** | 30.46 | 33.72 | 1.10 |
| **blood_pressure** | 4.90 | 4.14 | 14.27 |
| **crp_level** | 1.25 | 1.00 | 3.60 |
| **homocysteine_level** | 1.09 | 0.97 | 3.42 |

These within-class standard deviations vary significantly across the diagnostic groups. For instance, in HT_HY, the variables systolic_blood_pressure, diastolic_blood_pressure, heart_rate, HDL, and LDL have extremely low variability relative to the classes DI and DI_HY. On the other hand, for the class HT_HY, blood_pressure, crp_level, and homocysteine_level have higher within-class

variability. The above could mean that the data was preprocessed differently from the sources, or some data imputation or generation was done; however, the provided data set cannot provide this information. The above observation is pertinent to LDA since the assumption of covariance sharing can be questionable in such a case.

## 3.6 Redundant, derived and potentially inconsistent predictors

Variable **calcium_baseline** is a constant with a value of 9.42. The variable **systolic_index** is an exact copy of the variable **systolic_bp**, while the variable **age_normalized** is a perfect linear transformation of **age** (r = 1.0). We hence did away with these three variables and kept the original variables **age** and **systolic_bp**. There are 30 original input columns in the model matrix.

Age_level and bmi_level are also part of the dataset as categorical encodings of age and bmi, respectively. Such encodings are redundant as measures but can possibly allow linear classifiers to express threshold relations and thus were kept as part of the main experiments. It should be noted that a consistency test discovered 115 observations where the value of low_hdl_cholesterol was equal to Yes despite the fact that the value of hdl was 40 or greater. It is therefore evident that the binary variable is not equivalent to the condition hdl < 40.

# 4 Experimental Design and Preprocessing

Splitting was performed only once, using fixed random seed 26 and stratification by target_class: 1,600 samples for training (80%), 400 samples for hold-out (20%). Class counts for training samples: DI_HY = 1,030, DI = 515, HT_HY = 55; class counts for test samples: DI_HY = 257, DI = 129, HT_HY = 14.

Both prediction tasks share the same patient split. However, none of the targets is used as a feature for the other task.

The preprocessing was incorporated in scikit-learn pipelines. Numerical features were median-imputed and standardized; categorical features were mode-imputed and one-hot encoded. The provided dataset does not have any missing values; however, imputation makes the pipeline reusable. Preprocessing fitting inside each validation fold prevents information leak.

Cross-validation on training was used for choosing a candidate approach; hold-out sample was used for descriptive evaluation only.

In the regression setting we present mean absolute error (MAE), root mean squared error (RMSE) and coefficient of determination ($R^2$). Lower MAE/RMSE and higher $R^2$ correspond to better scores; test $R^2$ may be negative. In the classification setting we focus on macro-F1 (meaning class-wise F1), balanced accuracy (class recall), accuracy and confusion matrices.

In order to study the impact of class imbalance, the model of Logistic Regression was estimated using equal class weights whereas LDA was tested using equal class priors. Empirical-priors LDA was also performed in order to test sensitivity to choice of prior distribution. Clearly, since the former two approaches deal with the problem of class imbalance in different ways, they are not able to capture an algorithmic-only effect. In order to study the effect of regularization, C-values were selected via stratified five-fold cross-validation on the training partition, where feature selection was included in each fold. The independent set of 400 cases was held out for validation purposes and was not used to tune the models.

# 5 Simple Linear Regression

## 5.1 Approach

This project asks for a model that is based on just one feature and a process that searches for the best such feature [1]. For every possible numerical feature, we constructed an OLS regression in isolation. The five-fold cross-validation on the training set evaluated each of these models, and the one with lowest mean RMSE on the validation sets was selected.

## 5.2 Results and interpretation

***Table 5 – Predictor Results and Interpretation***

| Predictor | 5-fold CV RMSE | 5-fold CV $R^2$ |
|---|---|---|
| **bmi** | 43.0517 | −0.0009 |
| **HbA1c_level** | 43.0755 | −0.0021 |
| **crp_level** | 43.1149 | −0.0037 |
| **hdl** | 43.1162 | −0.0037 |

BMI ranked first, although all cross-validation $R^2$ values listed above are approximately zero or negative. The fitted regression equation, expressed on the original BMI scale, is:

**predicted target_regression = 127.585 + 0.3926 × bmi**

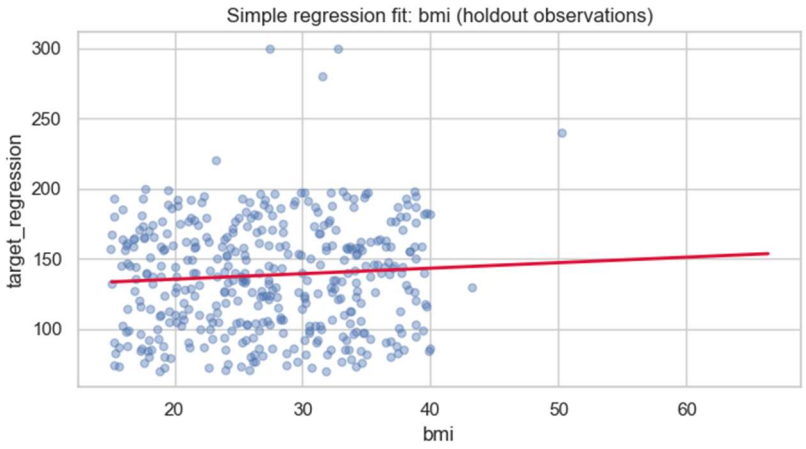



***Figure 6 - BMI-based simple linear fit and the independent test observations (notebook cell 20).***

*Table 6 - Holdout metrics*

| Holdout metric | BMI-only linear regression |
|---|---|
| MAE | 32.1912 |
| RMSE | 38.8853 |
| $R^2$ | 0.0032 |

With regard to the value of holdout $R^2$ being slightly positive, we see that it explains a bit less than 1% of variation relative to a constant prediction. The slope represents the estimated average relationship of 0.3926 target units for each BMI unit on the basis of a fitted linear model, not a causal relationship.

# 6 Multiple Linear Regression

## 6.1 Approach and feature-set comparison

In order to find out whether multiple clinical data would help in improving the prediction model's accuracy, five different models were compared:

- The top 2, 4, and 8 most correlated with the training fold clinical features;
- The fixed set of nine variables;
- All 30 non-redundant predictors (including categorical encoding where needed).

The ranking was done within the validation/training folds to avoid leakage from feature selection.

***Table 7 - Approach and feature-set comparison***

| Feature set | Variables | CV RMSE | CV $R^2$ |
|---|---|---|---|
| **Top 2 numeric** | 2 | 43.0421 | −0.0007 |
| **Top 4 numeric** | 4 | 43.1368 | −0.0053 |
| **Clinical indicators** | 9 | 43.1421 | −0.0057 |
| **Top 8 numeric** | 8 | 43.1785 | −0.0076 |
| **All cleaned features** | 30 | 43.3118 | −0.0142 |

The configuration selected utilized BMI and HbA1c level. The all-feature model performed worse overall in training cross-validation, suggesting the hypothesis that introducing more weak features increases estimation variance without increasing information.

## 6.2 Holdout regression comparison

***Table 8 - Holdout regression comparison***

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| **Training-mean reference** | 32.2331 | 38.9979 | −0.0026 |
| **Simple: BMI** | 32.1912 | 38.8853 | 0.0032 |
| **Multiple: BMI + HbA1c** | 32.1896 | 38.8743 | 0.0037 |

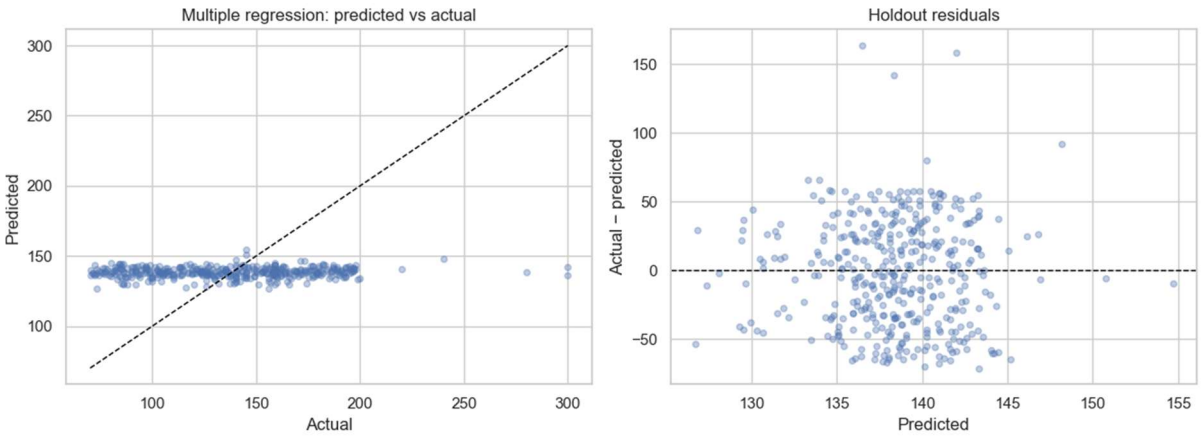



***Figure 7 - Predicted-versus-observed response and residuals for the selected multiple model (notebook cell 23).***

The RMSE for the multiple model (38.8743) is the smallest compared to that for the simple model (38.8853) and training mean reference (38.9979); however, its difference with the other two is very minimal. The value of $R^2$ is still almost equal to zero at 0.0037. Therefore, based on the results of the experiment, the target is hard to be predicted linearly from the given characteristics.

# 7 Classification: Logistic Regression versus LDA

## 7.1 Approach

We explored the diagnosis prediction task of three classes through two linear classifiers and a majority-class baseline. The Logistic Regression classifier was trained with class weighting in order to combat the issue of fewer instances of the HT_HY class. LDA employed pooled covariance with automatic shrinkage in order to mitigate correlation among features and uses class priors by default, contrary to the weighted logistic classifier. The model was chosen based on five-fold cross-validation training and macro-F1 scoring.

***Table 9 - Algorithm Performance***

| Algorithm | 5-fold macro-F1 | SD | Bal. accuracy | Accuracy |
|---|---|---|---|---|
| **LDA (shrinkage)** | 0.5997 | 0.0199 | 0.6641 | 0.5606 |
| **Logistic Regression** | 0.5768 | 0.0153 | 0.6382 | 0.5544 |
| **Majority baseline** | 0.2611 | 0.0000 | 0.3333 | 0.6438 |

The LDA model outperformed Logistic Regression based on the pre-specified F1 metric criterion for training-CV by around 0.0092; however, the latter performed better in terms of balanced accuracy. Such a gap can be considered rather small compared to the variation between folds.

## 7.2 Independent holdout results

***Table 10 - Independent holdout results***

| Algorithm | Accuracy | Balanced accuracy | Macro-F1 |
|---|---|---|---|
| **Logistic Regression** | 0.5150 | 0.5406 | 0.5176 |
| **LDA (uniform priors)** | 0.5100 | 0.5431 | 0.5240 |
| **Majority baseline** | 0.6425 | 0.3333 | 0.2608 |

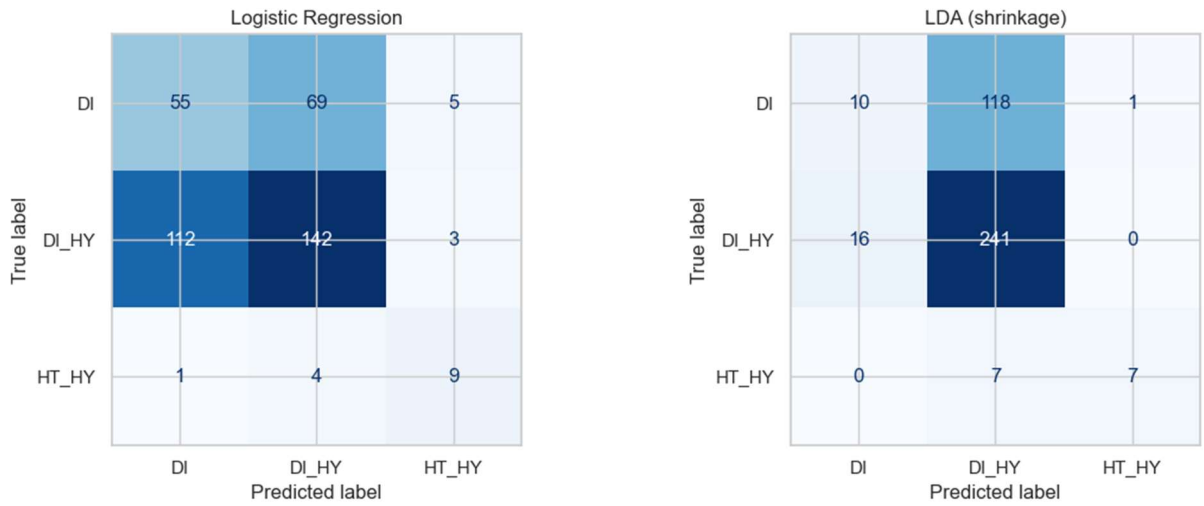



***Figure 8 - Three-class holdout confusion matrices (rows: true classes; columns: predictions; notebook cell 27).***

***Table 11 - Per-class holdout recall***

| True diagnosis | Logistic recall | LDA recall | Holdout support |
|---|---|---|---|
| **DI** | 0.426 | 0.457 | 129 |
| **DI_HY** | 0.553 | 0.529 | 257 |
| **HT_HY** | 0.643 | 0.643 | 14 |

Uniform-Prior LDA produced the best performance on training CV macro-F1 score (0.5997), narrowly beating balanced Logistic Regression (0.5768). It got the holdout CV macro-F1 of 0.5240, whereas Logistic Regression managed 0.5176. Uniform-Prior LDA got the recall of 0.457 for the DI class and 0.643 for the HT_HY class, with accuracy of 0.510. Majority Class produced better accuracy (0.6425) than the selected model, but considerably worse macro-F1 (0.2608), illustrating that accuracy alone is insufficient for an imbalanced problem. The chosen model was Uniform-Prior LDA, by means of predefined training CV macro-F1 criterion.

## 7.3 Sensitivity to LDA class priors

***Table 12 - Sensitivity to LDA class priors results***

| LDA configuration | Train CV macro-F1 | Holdout macro-F1 | DI recall | HT_HY recall |
|---|---|---|---|---|
| **Empirical priors** | 0.5860 | 0.5130 | 0.078 | 0.500 |
| **Uniform priors** | 0.5997 | 0.5240 | 0.457 | 0.643 |

Prior-sensitivity experimentation showed that the prediction by LDA was highly sensitive to class frequencies. Empirical-prior LDA provided a holdout accuracy score of 0.645 and identified only 7.8% of DI cases. Equal prior probability for the three classes led to an increase in the recall of DI cases to 45.7% and HT_HY recall to 64.3% with a simultaneous improvement of holdout macro-F1 from 0.5130 to 0.5240. Uniform-prior LDA is the best combination based on the predefined training-CV macro-F1 criterion. However, class prior setting does not eliminate the common-covariance assumption or attain clinical validity.

# 8 Resampling and Variance Analysis

## 8.1 Approach and measures

The classification algorithms were evaluated through stratified 5 and 10-fold cross-validation, 15 repetitions of a 25% stratified holdout split, Leave-One-Out Cross-Validation (LOOCV) for all 1,600 training instances, and 200 bootstraps with out-of-bag evaluation. The 400-instance independent holdout set was reserved separately. LOOCV resulted in one out-of-fold prediction for each training instance and the macro-F1 score was computed based on the aggregated predictions.

## 8.2 Repeated assessment results

***Table 13 - Repeated assessment results***

| Model / procedure | Repeats | Mean macro-F1 | Score SD | Score variance |
|---|---|---|---|---|
| **Logistic · 5-fold CV** | 5 | 0.57685 | 0.01530 | 0.00023 |
| **Logistic · 10-fold CV** | 10 | 0.57597 | 0.05878 | 0.00346 |
| **Logistic · repeated holdout** | 15 | 0.57270 | 0.02454 | 0.00060 |
| **Logistic · bootstrap OOB** | 200 | 0.57263 | 0.02926 | 0.00086 |
| **LDA · 5-fold CV** | 5 | 0.59967 | 0.01987 | 0.00040 |
| **LDA · 10-fold CV** | 10 | 0.57108 | 0.04414 | 0.00195 |
| **LDA · repeated holdout** | 15 | 0.57810 | 0.02809 | 0.00079 |
| **LDA · bootstrap OOB** | 200 | 0.57316 | 0.02626 | 0.00069 |

The benefit of LDA established using 5-fold cross-validation could not be proved using the other resampling techniques. The mean macro-F1 score of LDA was equal to 0.5997 for 5-fold CV, but it was 0.5711 for 10-fold CV, 0.5781 for repeated holdout, 0.5732 for bootstrap OOB and 0.5758 for LOOCV. Logistic Regression demonstrated greater consistency with the values between 0.5726 and 0.5769. Summing up all the methods, it is obvious that the two models show approximately equal

performance. The choice of LDA as the model under consideration was due to the definition of 5-fold training macro-F1 as the criterion.

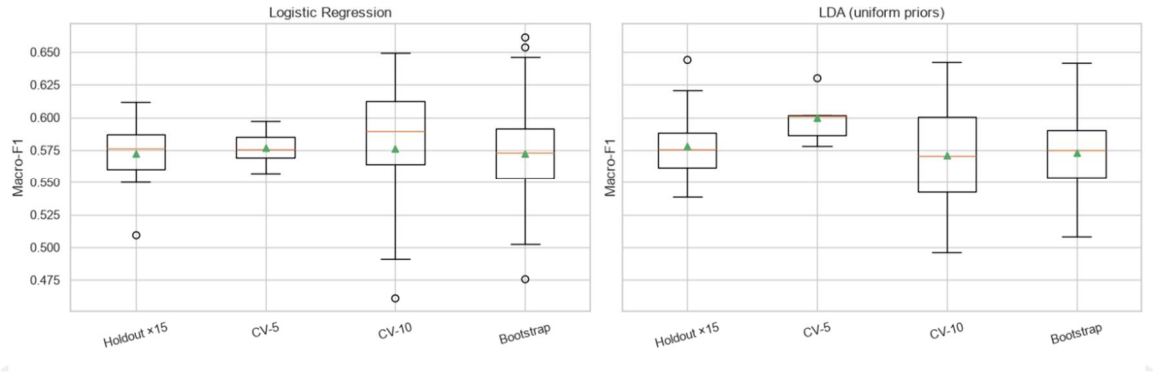



***Figure 9 - Distribution of macro-F1 scores across resampling procedures (notebook cell 36).***

The calculated value of the macro-F1 and its variability depending on the used method of evaluation was different. For Logistic Regression, the score SD in 5-fold CV was 0.01530, while the score SD in 10-fold CV was 0.05878. For the uniform-prior LDA, these SD values were 0.01987 and 0.04414 respectively. Bootstrap OOB means were 0.57263 and 0.57316 for Logistic Regression and LDA respectively. This variability is expressed in terms of SD and shows score variation depending on the test set size and dependent sampling.

## 8.3 LOOCV and bootstrap caveats

***Table 14 - LOOCV and bootstrap caveats***

| Model | Pooled macro-F1 | Balanced accuracy | Accuracy |
|---|---|---|---|
| **Logistic Regression** | 0.5761 | 0.6510 | 0.5575 |
| **LDA — uniform priors** | 0.5758 | 0.6544 | 0.5488 |

The full LOOCV generated almost the same pooled macro F1 score results of 0.5761 for Logistic Regression and 0.5758 for LDA under uniform priors. Pooled estimates using LOOCV method were generated by 1,600 individual fits per classifier. It follows that the ranking was not the same across different evaluation methods, and that the difference in the LOOCV results is not significant enough to claim any superiority.

The bootstrap experiment involved sampling 200 times of size 1,600 from the training set, with replacement. In both cases, all 200 samples were usable and no sample was omitted; and the average out-of-bag sample size was 589.2. Means and variances using bootstrapping methods provide only descriptive information about sampling sensitivity of the performance estimator and not validation.

# 9 Regularization and Classification Feature Selection

## 9.1 Approach

The assignment defines Ridge and Lasso methods in the framework of classification feature selection [1]. Ridge and Lasso classifiers were chosen to be Logistic Regression with L2 penalty as the Ridge method, logistic regression with L1 penalty as one versus rest Lasso model, and logistic regression with L2 penalty applied after L1-based feature selection. The mentioned models represent classifiers' equivalents of L1/L2 norms in Ridge and Lasso regularization, rather than regular ridge and lasso regressions for continuous outcome variable. Model feature selector is fitted in the folds separately to prevent leakage.

In order to choose the optimal value of the regularization strength, we performed stratified five-fold cross-validation on the training dataset for different values of the inverse regularization parameter C. In order to assess the performance of the model, we used macro-F1 scoring, such that every diagnostic category had equal influence on the results.

As for the small model, L1-based feature selection was added to the preprocessing step and was done separately for each fold of cross-validation training set in order to avoid data leakage. The maximum number of selected encoded features was set to 20. We aimed to examine the impact of a smaller number of predictors on the performance of the classifier.

This connection between the penalty and the number of selected columns proves that the size of the reduced model depends mostly on this threshold. When C was set to 0.03, the L1 selector used only 6 encoded columns, and when it was 0.05, 19 columns. After that, starting from 0.08, the selector selected the maximum number of 20 columns. In this threshold, changing the value of C affects only which coefficients are selected, but not the amount of features used, thus this chosen value of C (0.3) should be considered as a tuning of the selector with a threshold of 20 columns.

There are still 30 predictors left after structural cleaning, but the process of one-hot encoding gives 40 predictors for model. Reduced model selection mask is applied to 40 predictors after encoding; it does not mean that there are only 20 original clinical predictors.

## 9.2 Validation and holdout results

| Feature selection summary | Value |
|---|---|
| Original input predictors after structural exclusions | 30 |
| Columns after one-hot encoding | 40 |
| Reduced model selected encoded columns | 20 |
| Encoded columns removed by reduced model | 20 |
| Nonzero columns in full OvR L1 model | 40 of 40 |
| Training-CV-selected regularized configuration | Full Logistic L2 |

***Table 15 - Regularization model comparison***

| Regularization configuration | CV macro-F1 | CV SD | Chosen C | Holdout macro-F1 |
|---|---|---|---|---|
| Full Logistic L2 | 0.5848 | 0.0161 | 0.6 | 0.5212 |
| Full OvR Logistic L1 | 0.5827 | 0.0101 | 2.0 | 0.5283 |
| Reduced L1 selection + Logistic L2 | 0.5796 | 0.0256 | 0.3 | 0.5430 |

***Table 16 - Feature selection summary***

Regularization experiments involved three classifier configurations, which included full Logistic Regression with L2 regularization, full one-versus-rest Logistic Regression with L1 regularization, and reduced number of features through a combination of L1 feature selection and L2 Logistic Regression.

The full L2 classifier had the best training cross-validation macro-F1 score (0.5848), where C=0.6. Full OvR classifier with L1 regularization had similar macro-F1 score of 0.5827 at C=2.0, whereas the reduced classifier had a macro-F1 score of 0.5796 at selected feature selection parameter of C=0.3. It can be concluded that all three configurations had similar performance on validation set with slight preference towards full L2 configuration.

In the case of feature selection, full L1 classifier had nonzero coefficients for all the 40 encoded predictors, which implies that sparse solution will not always be obtained even with L1 regularization if a weaker selected penalty is applied. On the other hand, reduced feature set included only 20 encoded predictors, hence reducing input dimension by 50%.

In the holdout set that was not used for learning, the reduced model performed better by attaining a higher value of macro-F1 of 0.5430 in contrast to 0.5283 in the full L1 model and 0.5212 in the full L2 model. Nevertheless, the reduced model did not get the highest score of cross-validation in the training phase. Consequently, its good performance in the holdout test is a fact that does not give any grounds to change the selection criterion retrospectively.

In summary, the above-mentioned experiment shows that feature reduction could keep the prediction ability, even though feature reduction does not necessarily mean good generalization. Under the given cross-validation criterion, the full L2 Logistic Regression is still considered the preferable choice in the regularization case.

# 10 Critical Discussion and Study Limitations

The experimental outcomes clearly demonstrate the considerable difference between the two supervised learning tasks. In terms of prediction, the regression models showed very weak predictive power, where the best model was multiple linear regression with holdout $R^2 = 0.0037$. At the same time, the classification task displayed observable patterns; however, class imbalances and model assumptions had a significant impact on the evaluation results. For instance, LDA with equal class priors provided the highest five-fold training cross-validation macro-F1 (0.5997) as opposed to balanced Logistic Regression's 0.5768. Still, in terms of holdout macro-F1 (0.5240 and 0.5176, respectively), the models demonstrated rather poor generalization capabilities with LDA having the edge.

Several potential sources of bias have been deliberately avoided in the experimental setup. Both the classification and regression tasks utilize a single, stratified hold-out dataset that has never been involved in the selection of models or parameters. Preprocessing, as well as feature selection and ranking, have been done within the folds of cross-validation, thereby eliminating the possibility of validation data influencing training. Each task has also been evaluated against a trivial baseline, allowing us to tell whether the model is learning something other than the frequency of classes or the mean of the target variable.

Nonetheless, the following issues need to be taken into account:

- Limited representation in minority classes: In the HT_HY class there are only 69 examples in the whole dataset and 14 in the independent holdout set. This implies that class-specific recall estimations depend on a few instances which are either misclassified or classified correctly.
- Dependence between estimation values during resampling: The standard deviation values for performance measures come from the validation process where different sample sizes were used in training and evaluating stages. These standard deviation values are not comparable measures of variance in model predictions as they correspond to different resampling procedures.

- Regularization approach: The L1- and L2-regularized logistic regression is a regularization method applied to a classification error measure, while Lasso and Ridge regression use the least squares loss function to regularize. Despite having similar penalties these optimization problems are not equivalent.
- Limited external validity: Patient data are collected at one point in time and do not include an independent externally collected dataset which can be separated in terms of time or institution. Thus, it is impossible to demonstrate external validity or reliability of the predictions made.

A further aspect of the same problem is associated with the notable heterogeneity in the class-conditional variability of features detected in the exploratory analysis of the dataset. For example, there were several features (systolic blood pressure, diastolic blood pressure, heart rate), which had notably smaller standard deviations in HT_HY as compared to other classes, as well as features (CRP and homocysteine) whose variability in HT_HY exceeded their variability in other classes. These observations could have occurred due to some heterogeneity of the patients included into HT_HY or the specificities of data collection and processing, potentially even involving imputation artefacts. The available data does not allow to trace their origin.

The problem is particularly important for LDA due to the assumption about the common covariance matrix within the class. While covariance shrinkage helps to solve the problems with numerical instability and parameter estimation, it does not ensure the validity of the mentioned assumption.

Additionally, the regularization trials optimized the hyperparameters using cross-validation and the cross-validation scores of the selected parameter configurations have been derived using the same folds that were used during tuning, which may lead to optimism in the cross-validation scores. Independent holdout scores offer a separate evaluation mechanism, but the latter is still prone to uncertainty due to the small number of samples from the minority class.

Further research should focus on the origin and preprocessing of the clinical measurements, using nested cross-validation to estimate the performance of model selection, uncertainty intervals for macro-F1 and class-based recall and external validation using independently obtained patient data. Furthermore, additional classification algorithms with other distributional assumptions can be examined to evaluate the weaknesses of LDA.

# 11 Conclusion

Our team considered exploratory data analysis, simple and multiple linear regression, classification, resampling and regularization-based feature selection using the dataset of 2,000 patient records.

The regression analysis was unable to show any explanatory power, where the best multiple linear regression produced a holdout $R^2$ of 0.0037. For the classification analysis, the best five-fold training cross-validation macro-F1 results were achieved by Linear Discriminant Analysis (LDA) with equal class prior probabilities (0.5997). This is higher than the balance logistic regression macro-F1 value of 0.5768. The selected LDA pipeline produced a holdout macro-F1 score of 0.5240.

Leave-one-out cross-validation with 1,600 training observations resulted in very close pooled macro-F1 values for both classifiers. Additionally, 200 bootstrap replicates provided further information regarding the variability in the evaluation process.

For the regularization analysis, the full L2-regularized logistic regression model showed the best training cross-validation results. However, the reduced feature model was able to preserve only 20 out of 40 encoded features but still maintain competitive holdout results.

Finally, it was found that there is considerable variation in the within-class variance of features, especially in the case of HT_HY diagnostic category. Coupled with class imbalance, the above findings raise issues regarding model assumptions and generalization, which must be considered carefully when interpreting the findings.

All in all, this research highlights the significance of setting up criteria prior to comparing the models, avoiding the phenomenon of data leakage by using the right pre-processing pipelines, and interpreting the results according to the nature of the data set and assumptions of the methodology. It's concluded that the associations identified in this study are predictive, not causal, and do not establish actual clinical utility.

# 12 References

[1] Instituto Superior de Engenharia do Porto. (2026/2027). Trabalho_APRAU_2026_27_step1: Aprendizagem Automática assignment (version 1.0), Sections 2–6. Course handout.

[2] Gomes, E. F. (2026/2027). Linear Regression; Classification — Logistic Regression & Discriminant Analysis; Evaluation & Resampling; Linear Model Selection & Regularization. APRAU lecture slides, ISEP.

[3] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). An Introduction to Statistical Learning: With Applications in Python. Springer. https://doi.org/10.1007/978-3-031-38747-0

[4] scikit-learn developers. (n.d.). scikit-learn user guide and API documentation. https://scikit-learn.org/stable/ (software version observed in notebook: 1.9.1).

[5] Group 26. (2026). APRAU_Group26.ipynb and dataset_grupo_26.csv. Executed analysis notebook and group-assigned dataset (primary sources for all report figures, tables and performance values).

# 13 Appendix A. Reproducibility and Submission Notes

- **Notebook:** APRAU_Group26.ipynb. The notebook comprises 43 cells, which include 24 executed code cells and figures.
- **Environment:** Python 3.13.9 and scikit-learn 1.9.1.
- **Seed:** 26.
- **Holdout experimental split:** 80/20 stratified.

All numerical results presented in this Word document were typed from the saved notebook results.